

Jacob Higgins

Aerospace Engineering Masters Student

Bristol, United Kingdom | jacob.higgins.2021@bristol.ac.uk | +447539352161

Education

University of Bristol

Aerospace Engineering, MEng | On track for 1st class honours

Bristol, UK

Graduation Date: 06/07/2025

Bishop Vesey's Grammar School

A Levels: Maths, Further Maths, Physics, Chemistry | Grades: A*A*A*A

Birmingham, UK

Graduated: 07/2021

GCSEs: 10, incl. Maths & English | Grades: 9-7

Roles

University of Bristol Satellite Ground Station

June 2023 - Present

Lead Engineer

- Secured a £10,000 grant to upgrade ground station facilities.
- Led procurement and construction efforts, collaborating with international manufacturers.
- Partnered with industry leaders to offer facility services to commercial entities for the first time.
- Diagnosed and resolved system hardware faults, enabling the reception of over 1,000 satellite transmissions.
- Achieved the first successful spacecraft transmission from the facility.
- Collaborated with team members and academics on design decisions at regular stand-up meetings.

Space Universities Network

February 2025 - Present

Teaching Associate

- Aided in developing educational short courses on rocket safety in collaboration with university academics.
- Reviewed and refined case study resources for UK higher education institutions, collaborating with industry partners to analyse future space concepts.
- Aided in creating resources and tracking engagement for monthly meetings between Space Universities Network and the UK Space Agency.
- Organized quarterly meetings for academics teaching space-related subjects in the UK.
- Contributed to the creation of monthly newsletters distributed to network members.

UoBSat, University of Bristol

September 2024 - Present

Lead Systems Engineer

- Led weekly concurrent design sessions with 15+ engineers.
- Delivered six educational sessions on space systems design processes and standards.
- Collaborated with sub-team leads to define mission requirements, guiding the team through a successful Requirements Readiness Review and Preliminary Design Review.

PROVE-Pathfinder, University of Bristol

September 2023 - Present

Thermal Lead

- Worked with team members to provide twice-annual outreach events, teaching local 16-18 year old students classes about satellite design.
- Conducted experimental testing on a satellite payload, diagnosing the root cause of previous failures.
- Developed a thermal model of the payload, which was later used for its qualification for a launch opportunity.
- Strengthened technical writing and communication skills through a testing report, and presenting results to project stakeholders.

Experiences

European Space Agency CubeSat Summer School 2024

1-28 August 2024

- Selected as one of 30 students from across Europe to attend ESA's CubeSat summer school in person at ESEC, Belgium.
- Worked in a 6-person international team to develop a satellite mission business case, presenting it to an audience of 50+, including ESA Business Incubation Centre representatives at the Euro Space Centre.
- Led a condensed environmental testing campaign for a satellite payload and presented a detailed test report to an ESA panel.
- Completed an intensive four-week program featuring 40+ lectures on spacecraft and rocket design, space sustainability, space law, and the space industry.

Mach24 Design Competition, BristolSEDS

November 2023 - July 2024

- Achieved first place in the rocketry and combined categories.
- Evaluated the bandwidth and baud rate of the payload's communication system using a software defined radio, to ensure compliance with Ofcom regulation 2210, and assist in the application of an operational safety case.
- Created a simulation tool to determine the longitudinal stability and centre of gravity requirement of the payload - a 2m folding wingspan drone, to be launched from a 1m rocket payload bay.

Summer Time Advanced Aerospace Residency, RAF Cosford

23-28 July 2018

Aerospace Intern with Northrop Grumman

- Assembled and presented a mission brief to over 30 people, including Northrop Grumman's International communications manager.
- Improved practical skills during activities including operating radar dishes, using fibre optic cameras to identify fractures inside engines, and measuring lift and drag of wings inside a wind tunnel.